

Reviewer Pack — Minimal Rank of Configurable Sheaves vs SOS Degree for Max-C...

Entry 34e93397354d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Configurable Sheaves vs SOS Degree for Max-CUT

Entry ID: 34e93397354d **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 19:28:06 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Configurable Sheaves) **Field B** (complexity object): Complexity Theory: Sum-of-squares Lower Bounds for Max-CUT

Statement:

For any instance of max-CUT, if its SOS degree is less than d , then the minimal rank of its corresponding configurable sheaf is at most d .

Rationale (proposer’s reasoning):

Configurable sheaves are a generalization of tropical geometry that allows for a more refined analysis of polynomials. This conjecture suggests that the structure captured by configurable sheaves could provide insights into the SOS degree of max-CUT, potentially leading to improved lower bounds in complexity theory.

Taxonomy category: SOS_DEGREE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): fd19a376300bd389

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

If the SOS degree is less than d , then the minimal rank of the configurable sheaf for all max-CUT instances is at most d .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 4 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "configurable sheaves" AND "SOS degree" AND "Max-CUT" - "minimal rank" AND "algebraic geometry" AND "sum-of-squares lower bounds" - "SOS degree" IN "complexity theory" AND "configurable sheaf"

Top relevant hits considered: - [http://arxiv.org/abs/2503.14756v3] SceneEval: Evaluating Semantic Coherence in Text-Conditioned 3D Indoor Scene Synthesis - [http://arxiv.org/abs/2408.02211v2] SceneMotifCoder: Example-driven Visual Program Learning for Generating 3D Object Arrangements - [http://arxiv.org/abs/2509.13291v1] Towards an Embodied Composition Framework for Organizing Immersive Computational Notebooks - [s2:9c7ab07901788d817d7832ba794199a58ad8bae8] Discrete and computational geometry : the Goodman-Pollack Festschrift

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_max_cut_instance(n):
        edges = []
        for i in range(n):
            for j in range(i + 1, n):
                if random.random() < 0.5:
                    edges.append((i, j))
        return edges

    def sos_degree(instance):
        # Placeholder implementation of SOS degree calculation
        return len(instance)

    def minimal_rank(instance):
        # Placeholder implementation of minimal rank calculation
        return len(instance)

    n = random.randint(5, 40) # Sweep through different sizes
    instance = generate_max_cut_instance(n)
    sos_d = sos_degree(instance)
    rank = minimal_rank(instance)

    metric_name = "SOS Degree vs Minimal Rank"
    metric_value = sos_d / rank if rank != 0 else float('inf')
```


8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The conjecture has been tested on only one instance (n=1), which is insufficient to draw a general conclusion. The metric does not scale with n, and the result may be an artifact of the specific case chosen.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The conjecture was tested on only one instance, which is insufficient to draw a general conclusion. The critic challenged the result due to the lack of scalability with respect to the number of instances. | next: Test the conjecture on a larger and more diverse set of max-CUT instances to verify its validity across different cases.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	9782
2	preregistration	ollama_remote	glm4:latest	0	0	5234
3	novelty	ollama_remote	glm4:latest	0	0	4769
4	novelty	ollama_remote	glm4:latest	0	0	5885
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12223
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8209
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6898
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7354
9	critic	ollama_remote	glm4:latest	0	0	9917
10	judge	ollama_remote	glm4:latest	0	0	5476

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 75748 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/34e93397354d.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/34e93397354d.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/34e93397354d.tar.gz (if generated)